Lecture 13

Finite-dimensional existence and classification

Objectives. *For each $\lambda \in P_+$, an explicit quotient of $M(\lambda)$ is finite-dimensional and simple. This completes the classification begun in Lecture 12.*

By the end of this lecture you will be able to:

- Write the quotient presentation of $L(\lambda)$ for $\lambda \in P_+$.
- Prove local finiteness and Weyl invariance for the simple-root $\mathfrak{sl}_2$ actions.
- Prove that the quotient is finite-dimensional and isomorphic to $L(\lambda)$.
- Prove the bijection between P_+ and the isomorphism classes of finite-dimensional simple modules.

Throughout, $\mathfrak{g}$ is a complex semisimple Lie algebra with Cartan subalgebra $\mathfrak{h}$, simple roots $\alpha_1, \dots, \alpha_\ell$ and Weyl group W, as in Lecture 9. For $\mathfrak{g} \neq 0$, Verma modules are infinite-dimensional.

13.1 The quotient presentation

For each i fix $e_i \in \mathfrak{g}_{\alpha_i}$ and $f_i \in \mathfrak{g}_{-\alpha_i}$ with $h_i = [e_i, f_i] = \alpha_i^\vee$. Then

$$[h_i, e_i] = 2e_i, \qquad [h_i, f_i] = -2f_i, \qquad [e_i, f_i] = h_i.$$

The span $\mathfrak{s}_i$ of e_i, h_i and f_i is a copy of $\mathfrak{sl}_2$ (Lecture 8). Write

$$Q_+ = \sum_{i=1}^{\ell} \mathbb{Z}_{\geq 0}\alpha_i, \qquad \mu \leq \lambda \iff \lambda - \mu \in Q_+.$$

A weight μ lies in P exactly when every $\mu(h_i)$ is an integer. The simple reflections act by

$$s_i\mu = \mu - \mu(h_i)\alpha_i.$$

We use the following facts from earlier lectures.

(F1) For $i \neq j$, $\alpha_j - \alpha_i$ is not a root, so $[e_j, f_i] = 0$. The e_i generate $\mathfrak{n}_+$. To see this, write a nonsimple positive root as $\beta = \gamma + \delta$ with $\gamma, \delta \in R_+$ (Lecture 9). Then $[\mathfrak{g}_\gamma, \mathfrak{g}_\delta] = \mathfrak{g}_\beta$ by Lecture 8. Induction on height shows that each $\mathfrak{g}_\beta$ with $\beta \in R_+$ lies in the subalgebra generated by the e_i.

(F2) $M(\lambda) = U(\mathfrak{n}_-)v_\lambda$ has weights in $\lambda - Q_+$, finite-dimensional weight spaces and $M(\lambda)_\lambda = \mathbb{C}v_\lambda$. PBW monomials applied to v_λ form a basis. If a module V is generated by $u \in V_\lambda$ with $\mathfrak{n}_+ u = 0$, then $v_\lambda \mapsto u$ extends to a surjection $M(\lambda) \to V$. The module $M(\lambda)$ has a unique maximal proper submodule, so every simple quotient is isomorphic to $L(\lambda)$ (Lecture 12).

(F3) W is finite and generated by the s_i. Every $\mu \in P$ is W-conjugate to an element of P_+. For $\lambda \in P_+$, the set $\{\mu \in P_+ : \mu \le \lambda\}$ is finite (Lecture 9).

(F4) Finite-dimensional modules of $\mathfrak{g}$, and of $\mathfrak{sl}_2$, are completely reducible (Lecture 5). The finite-dimensional simple $\mathfrak{sl}_2$-modules are the V_n of Lecture 6.

Lemma 13.1 (An $\mathfrak{sl}_2$ identity · *P*)**.** *In $U(\mathfrak{s}_i)$, for $k \ge 1$,*

$$e_i f_i^k = f_i^k e_i + k\, f_i^{k-1}(h_i - k + 1).$$

Proof. The relation $h_i f_i = f_i(h_i - 2)$ and induction on r give

$$h_i f_i^r = f_i^r (h_i - 2r) \qquad (r \ge 0).$$

Write the commutator as a telescoping sum. Then use $[e_i, f_i] = h_i$, the relation above with $r = k - 1 - j$, and $\sum_{j=0}^{k-1} 2(k-1-j) = k(k-1)$:

$$\begin{aligned} e_i f_i^k - f_i^k e_i &= \sum_{j=0}^{k-1} f_i^{\,j}\,[e_i, f_i]\, f_i^{\,k-1-j} \\ &= \sum_{j=0}^{k-1} f_i^{\,j} h_i f_i^{\,k-1-j} \\ &= \sum_{j=0}^{k-1} f_i^{k-1}\big(h_i - 2(k-1-j)\big) \\ &= k\, f_i^{k-1} h_i - k(k-1) f_i^{k-1} \\ &= k\, f_i^{k-1}(h_i - k + 1). \end{aligned}$$

∎

Definition 13.2 (The candidate module)**.** Let $\lambda \in P_+$ and $m_i = \lambda(h_i) \in \mathbb{Z}_{\ge 0}$. Put

$$N(\lambda) = \sum_{i=1}^{\ell} U(\mathfrak{g})\, f_i^{m_i+1} v_\lambda, \qquad V(\lambda) = M(\lambda)/N(\lambda).$$

Let u be the image of v_λ in $V(\lambda)$.

Lecture 12 used the symbol $N(\lambda)$ for the maximal proper submodule of $M(\lambda)$. In this lecture $N(\lambda)$ means the submodule just defined. Theorem 13.9 proves that the two submodules agree. Until then we may not assume that $V(\lambda)$ is $L(\lambda)$.

A *singular vector* is a nonzero weight vector w with $\mathfrak{n}_+ w = 0$.

The rest of the lecture follows a route, which we state here as a preview. The vectors $f_i^{m_i+1} v_\lambda$ generalise the vector v_{m+1} of the $\mathfrak{sl}_2$ example in Lecture 12. First, these vectors are singular, and they generate submodules below the top weight λ, so the quotient $V(\lambda)$ is nonzero. Next, every vector of $V(\lambda)$ is killed by a power of each e_i and each f_i. Finite-dimensional $\mathfrak{s}_i$-submodules then give Weyl symmetry of the weights. That symmetry and Lecture 9 leave finitely many weights, and PBW bounds the dimension of each weight space.

Finally, Weyl's theorem makes the cyclic quotient with a one-dimensional top weight space simple.

Proposition 13.3 (Singular vectors and a nonzero quotient · *P*). *Each* $w_i = f_i^{m_i+1} v_\lambda$ *satisfies* $\mathfrak{n}_+ w_i = 0$. *Moreover* $N(\lambda)_\lambda = 0$, *so* $u \neq 0$ *and* $V(\lambda)_\lambda = \mathbb{C}u$.

Proof. By PBW, w_i is a nonzero weight vector:

$$w_i \in M(\lambda)_{\lambda-(m_i+1)\alpha_i}.$$

For $j \neq i$, (F1) gives $e_j f_i = f_i e_j$, so

$$e_j w_i = f_i^{m_i+1} e_j v_\lambda = 0.$$

For $j = i$, Lemma 13.1 with $k = m_i + 1$ gives

$$\begin{aligned} e_i w_i &= f_i^{m_i+1} e_i v_\lambda + (m_i+1) f_i^{m_i} (h_i - m_i) v_\lambda \\ &= 0 + (m_i+1)(m_i - m_i) f_i^{m_i} v_\lambda = 0. \end{aligned}$$

The annihilator of w_i in $\mathfrak{g}$ is a subalgebra. It contains every e_j, so it contains $\mathfrak{n}_+$ by (F1).

PBW gives the triangular factorisation (Lecture 11)

$$U(\mathfrak{g}) = U(\mathfrak{n}_-)U(\mathfrak{h})U(\mathfrak{n}_+).$$

Here $\mathfrak{h}$ and $\mathfrak{n}_+$ preserve $\mathbb{C}w_i$. Hence

$$U(\mathfrak{g})w_i = U(\mathfrak{n}_-)U(\mathfrak{h})U(\mathfrak{n}_+)w_i = U(\mathfrak{n}_-)w_i.$$

This space is spanned by weight vectors whose weights μ satisfy

$$\lambda - \mu \in (m_i+1)\alpha_i + Q_+.$$

The coefficient of α_i in $\lambda - \mu$ is at least 1, so $\mu \neq \lambda$.

The λ-weight space of a sum of weight submodules is the sum of their λ-weight spaces. Thus

$$N(\lambda)_\lambda = \sum_{i=1}^{\ell} \big(U(\mathfrak{g})w_i\big)_\lambda = 0.$$

Hence $u \neq 0$, and $V(\lambda)_\lambda$ is the image $\mathbb{C}u$ of $M(\lambda)_\lambda = \mathbb{C}v_\lambda$. ∎

Since $N(\lambda)$ is spanned by weight vectors, $V(\lambda)$ is the direct sum of its weight spaces. These weight spaces are finite-dimensional by (F2). Since $\lambda \in P$, the weights lie in $\lambda - Q_+ \subseteq P$. The module $V(\lambda)$ is generated by u, and

$$\mathfrak{n}_+ u = 0, \qquad f_i^{m_i+1} u = 0 \quad (1 \le i \le \ell).$$

13.2 The simple-root $\mathfrak{sl}_2$ actions

Lemma 13.4 (Moving a power past x · *P*). *For* $x, y \in \mathfrak{g}$ *and* $N \ge 0$, *in* $U(\mathfrak{g})$,

$$y^N x = \sum_{j=0}^{N} \binom{N}{j} \big((\operatorname{ad} y)^j x\big)\, y^{N-j}.$$

Proof. Let L_y and R_y be left and right multiplication by y on $U(\mathfrak{g})$. Associativity and the definition of $\operatorname{ad} y$ give

$$L_y R_y = R_y L_y, \qquad \operatorname{ad} y = L_y - R_y.$$

Hence $\operatorname{ad} y$ commutes with R_y, and the binomial theorem applies:

$$L_y^N = (R_y + \operatorname{ad} y)^N = \sum_{j=0}^{N} \binom{N}{j} R_y^{N-j} (\operatorname{ad} y)^j.$$

Apply both sides to x. ∎

Proposition 13.5 (Local nilpotence · *P*)**.** *For each i and each $w \in V(\lambda)$, there is k with $e_i^k w = 0$ and $f_i^k w = 0$.*

Proof. Let $y = e_i$ or $y = f_i$, and put $\mathfrak{g}_0 = \mathfrak{h}$. Then $\operatorname{ad} y$ maps $\mathfrak{g}_\beta$ into $\mathfrak{g}_{\beta+\alpha_i}$ or $\mathfrak{g}_{\beta-\alpha_i}$, respectively. Only finitely many β have $\mathfrak{g}_\beta \neq 0$, so there is r with

$$(\operatorname{ad} y)^r = 0 \quad \text{on } \mathfrak{g}.$$

Since $y^k w = 0$ implies $y^{k'} w = 0$ for $k' \geq k$, the set

$$V' = \{w \in V(\lambda) : y^k w = 0 \text{ for some } k \geq 0\}$$

is a subspace. Suppose $y^k w = 0$ and $x \in \mathfrak{g}$. Apply Lemma 13.4 with $N = k + r$. Each term has $j \geq r$, so $(\operatorname{ad} y)^j x = 0$, or $k + r - j \geq k$, so $y^{k+r-j} w = 0$. Hence

$$y^{k+r} x w = \sum_{j=0}^{k+r} \binom{k+r}{j} ((\operatorname{ad} y)^j x) y^{k+r-j} w = 0.$$

So $xw \in V'$, and V' is a submodule.

The submodule V' contains u. For $y = e_i$ this holds since $e_i u = 0$, and for $y = f_i$ since $f_i^{m_i+1} u = 0$. As u generates $V(\lambda)$, we get $V' = V(\lambda)$. Take the larger of these two exponents. ∎

Proposition 13.6 (Local finiteness · *P*)**.** *Every $w \in V(\lambda)$ lies in a finite-dimensional $\mathfrak{s}_i$-submodule spanned by weight vectors.*

Proof. Every w is a finite sum of weight vectors, and a finite sum of such submodules is again one. So we may assume that w has weight μ. By Proposition 13.5, choose A and B with

$$e_i^{B+1} w = 0, \qquad f_i^{A+1} e_i^b w = 0 \quad (0 \leq b \leq B).$$

Let

$$X = \operatorname{span}\{f_i^a e_i^b w : \ 0 \leq a \leq A, \ 0 \leq b \leq B\}.$$

Each spanning vector is a weight vector:

$$f_i^a e_i^b w \in V(\lambda)_{\mu + (b-a)\alpha_i}.$$

So X is $\mathfrak{h}$-stable. It is f_i-stable by the choice of A.

For stability under e_i, put $c = (\mu + b\alpha_i)(h_i)$, so $h_i e_i^b w = c\, e_i^b w$. For $a \geq 1$, Lemma 13.1 gives

$$\begin{aligned} e_i f_i^a e_i^b w &= f_i^a e_i^{b+1} w + a\, f_i^{a-1} (h_i - a + 1) e_i^b w \\ &= f_i^a e_i^{b+1} w + a(c - a + 1) f_i^{a-1} e_i^b w. \end{aligned}$$

Both terms lie in X, and the first is zero when $b = B$. For $a = 0$ the vector is $e_i^{b+1} w$, again in X or zero. Thus X is an $\mathfrak{s}_i$-submodule. ∎

Proposition 13.7 (Weyl invariance · *P*). *For every weight μ and every $\sigma \in W$,*

$$\dim V(\lambda)_{\sigma\mu} = \dim V(\lambda)_\mu.$$

Proof. Each s_i maps P to P, since $\alpha_i \in P$. So W permutes P. Every weight of $V(\lambda)$ lies in $\lambda - Q_+ \subseteq P$. If $\mu \notin P$, then $\sigma\mu \notin P$, and both weight spaces vanish. Let $\mu \in P$.

The s_i generate W by (F3), so it suffices to treat $\sigma = s_i$. Write $V = V(\lambda)$. Then

$$k = \mu(h_i) \in \mathbb{Z}, \qquad s_i\mu = \mu - k\alpha_i.$$

Replacing μ by $s_i\mu$ if necessary, assume $k \geq 0$. We show that

$$f_i^k : V_\mu \to V_{\mu - k\alpha_i}, \qquad e_i^k : V_{\mu - k\alpha_i} \to V_\mu$$

are injective.

Let $x \neq 0$ be a weight vector of weight μ or $\mu - k\alpha_i$. By Proposition 13.6, x lies in a finite-dimensional $\mathfrak{s}_i$-submodule X. By (F4),

$$X = X_1 \oplus \cdots \oplus X_s, \qquad X_t \cong V_{n_t}.$$

Write $x = x_1 + \cdots + x_s$ with $x_t \in X_t$. The projections onto the X_t commute with h_i. So each nonzero component x_t of x is an h_i-eigenvector with eigenvalue k or $-k$, respectively.

In V_n use the basis $v_j = f_i^j v_0$ of Lecture 6. It satisfies

$$\begin{aligned} h_i v_j &= (n - 2j)v_j, \\ f_i v_j &= v_{j+1} \quad (j < n), \\ e_i v_j &= j(n - j + 1)v_{j-1} \quad (j > 0). \end{aligned}$$

A k-eigenvector in V_n is a multiple of v_j with $n - 2j = k$. A $(-k)$-eigenvector is a multiple of $v_{j'}$ with $n - 2j' = -k$. In this case put $j = j' - k = n - j' \geq 0$. Then again $n - 2j = k$, and the $(-k)$-eigenvector is a multiple of v_{j+k}. In both cases $j + k = n - j \leq n$, so

$$\begin{aligned} f_i^k v_j &= v_{j+k} \neq 0, \\ e_i^k v_{j+k} &= \prod_{t=j+1}^{j+k} t(n - t + 1)\, v_j \neq 0, \end{aligned}$$

since $1 \leq t \leq n$ in the product. The vectors $f_i^k x_t$, or $e_i^k x_t$, are nonzero and lie in distinct summands. Hence $f_i^k x \neq 0$ in the first case and $e_i^k x \neq 0$ in the second. The two injections give

$$\dim V_\mu \leq \dim V_{s_i\mu} \leq \dim V_\mu.$$

■

13.3 Finite dimension and irreducibility

Lemma 13.8 (Cyclic modules · *P*). *Let V be a finite-dimensional $\mathfrak{g}$-module that is the direct sum of its weight spaces. Suppose $\dim V_\lambda = 1$ and V is generated by a vector $u \in V_\lambda$. Then V is simple.*

Proof. By (F4), $V = V^{(1)} \oplus \cdots \oplus V^{(s)}$ with simple submodules $V^{(t)}$. The projections onto the summands commute with $\mathfrak{h}$, so

$$V_\lambda = V^{(1)}_\lambda \oplus \cdots \oplus V^{(s)}_\lambda.$$

Since $\dim V_\lambda = 1$, exactly one summand, say $V^{(1)}$, has $V^{(1)}_\lambda \neq 0$. Then

$$u \in V_\lambda = V^{(1)}_\lambda, \qquad V = U(\mathfrak{g})u \subseteq V^{(1)}.$$

So $V = V^{(1)}$ is simple. ∎

Theorem 13.9 (Existence · *P*)**.** *Let $\lambda \in P_+$. The module $V(\lambda)$ is finite-dimensional, and $V(\lambda) \cong L(\lambda)$. In particular, $L(\lambda)$ is finite-dimensional, and $N(\lambda)$ is the maximal proper submodule of $M(\lambda)$.*

Proof. Finitely many weights. Let μ be a weight of $V(\lambda)$. Then $\mu \in P$, and (F3) gives $\sigma \in W$ with $\sigma\mu \in P_+$. By Proposition 13.7, $\sigma\mu$ is also a weight, so $\sigma\mu \leq \lambda$. Write $\operatorname{wt} V(\lambda)$ for the set of weights. Then

$$\operatorname{wt} V(\lambda) \subseteq \bigcup_{\nu \in P_+,\ \nu \leq \lambda} W\nu.$$

The index set is finite by (F3), and W is finite. So there are finitely many weights.

Finite dimension. Each weight space of $V(\lambda)$ is a quotient of a weight space of $M(\lambda)$. By (F2),

$$\dim V(\lambda) = \sum_{\mu \in \operatorname{wt} V(\lambda)} \dim V(\lambda)_\mu \leq \sum_{\mu \in \operatorname{wt} V(\lambda)} \dim M(\lambda)_\mu < \infty.$$

Irreducibility. By Proposition 13.3, Lemma 13.8 applies to $V(\lambda)$ and u. So $V(\lambda)$ is a simple quotient of $M(\lambda)$, and (F2) gives $V(\lambda) \cong L(\lambda)$. Its kernel $N(\lambda)$ is therefore the unique maximal proper submodule. ∎

13.4 The classification

Theorem 13.10 (Classification · *P*)**.** *The assignment $\lambda \mapsto L(\lambda)$ induces a bijection from P_+ to the set of isomorphism classes of finite-dimensional simple $\mathfrak{g}$-modules.*

Proof. Well defined. This is Theorem 13.9.

Surjective. By Lecture 12, every finite-dimensional simple module is isomorphic to $L(\lambda)$ for some $\lambda \in P_+$.

Injective. An isomorphism commutes with $\mathfrak{h}$, so it preserves weights. The weights of $L(\lambda)$ lie in $\lambda - Q_+$, and λ is one of them. If $L(\lambda) \cong L(\mu)$, then $\mu \in \lambda - Q_+$ and $\lambda \in \mu - Q_+$. Since the α_i are linearly independent,

$$\lambda - \mu \in Q_+ \cap (-Q_+) = \{0\}.$$

∎

We now look at the existence result for small modules of $\mathfrak{sl}_3$. The drawings use the plane of Lecture 9, in which the point (a, b) is the weight $a\omega_1 + b\omega_2$. For $\lambda = 2\omega_1$ we have $m_1 = 2$ and $m_2 = 0$, so the imposed vectors are $f_1^3 v_\lambda$ and $f_2 v_\lambda$. Figure 13.1 shows them inside the weights of $M(\lambda)$. The quotient does not simply remove the weights below these two vectors. For example, the weight $\lambda - 2\alpha_1 - \alpha_2$ has multiplicity 2 in $M(\lambda)$ and multiplicity 1 in $L(\lambda)$. Exercise 13.3 derives the six weights of $L(2\omega_1)$ and their multiplicities.

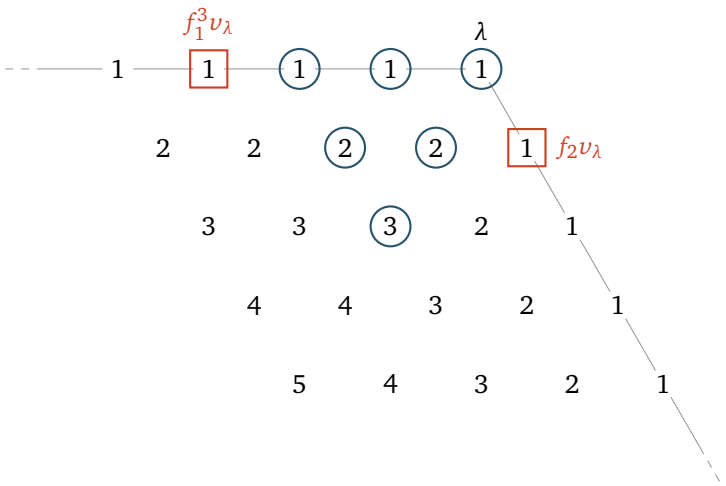


Figure 13.1: The weights of $M(2\omega_1)$ with their multiplicities, and the two imposed vectors in squares. Circles mark the six weights of $L(2\omega_1)$, each of multiplicity 1 there, so a circled number 2 or 3 drops to 1 in the quotient.

Proposition 13.7 says that multiplicities are constant on Weyl orbits. So the multiplicities at dominant weights determine the whole picture. Figure 13.2 shows this for four modules. Exercises 13.3 and 13.5 derive the first two panels. Lecture 17 gives a formula for multiplicities like those of the last two. They are printed here to show the pattern, and nothing below uses them.

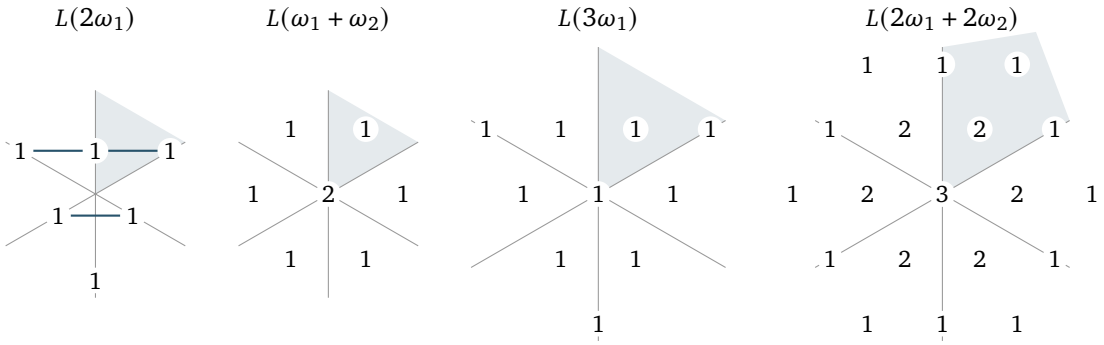


Figure 13.2: Weights of four simple $\mathfrak{sl}_3$-modules, each labelled by its multiplicity. The shaded region is part of the dominant chamber, and the grey lines are the reflecting lines. In the first panel the thick lines join the α_1-strings of three points and two points. The third string is the single point at the bottom.

Simple ideals

Decompose $\mathfrak{g}$ into simple ideals $\mathfrak{g}_k$ (Lecture 4). Choose a Cartan subalgebra $\mathfrak{h}_k \subseteq \mathfrak{g}_k$ and positive roots for each k. Write

$$\mathfrak{g} = \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_r, \qquad \mathfrak{h} = \mathfrak{h}_1 \oplus \cdots \oplus \mathfrak{h}_r.$$

This $\mathfrak{h}$ is toral, since $x = \sum_k x_k$ with $x_k \in \mathfrak{h}_k$ acts diagonalisably on $\mathfrak{g}_k$ by $\operatorname{ad} x_k$. It is also maximal. Indeed, a toral, hence abelian, subalgebra containing $\mathfrak{h}$ lies in

$$C_{\mathfrak{g}}(\mathfrak{h}) = \bigoplus_k C_{\mathfrak{g}_k}(\mathfrak{h}_k) = \mathfrak{h},$$

by the self-centraliser theorem [C] of Lecture 7 (Humphreys, §8.2), applied to each maximal toral $\mathfrak{h}_k$ of the semisimple $\mathfrak{g}_k$. So $\mathfrak{h}$ is a Cartan subalgebra of $\mathfrak{g}$, and we use it from now on.

For $j \neq k$, $\mathfrak{h}_j$ acts by zero on $\mathfrak{g}_k$. Hence the roots of $\mathfrak{g}$ are the roots of the $\mathfrak{g}_k$, extended by zero. Choose their union as the positive system of $\mathfrak{g}$. The simple roots and coroots of $\mathfrak{g}$ are the disjoint unions of those of the $\mathfrak{g}_k$, so

$$\mathfrak{h}^* = \mathfrak{h}_1^* \oplus \cdots \oplus \mathfrak{h}_r^*, \qquad P_+ = P_+^{(1)} \times \cdots \times P_+^{(r)}.$$

For $\mathfrak{g}_k$-modules (V_k, ρ_k), the *outer tensor product* $V_1 \boxtimes \cdots \boxtimes V_r$ is $V_1 \otimes \cdots \otimes V_r$ with $x = x_1 + \cdots + x_r$, with $x_k \in \mathfrak{g}_k$, acting by

$$\sum_{k=1}^{r} \mathrm{id} \otimes \cdots \otimes \rho_k(x_k) \otimes \cdots \otimes \mathrm{id}\,.$$

The terms for different k commute, and $[\mathfrak{g}_j, \mathfrak{g}_k] = 0$ for $j \neq k$. So this is a representation.

Theorem 13.11 (Factorisation · *P*). *For $\lambda = (\lambda^1, \dots, \lambda^r) \in P_+$,*

$$L(\lambda) \cong L(\lambda^1) \boxtimes \cdots \boxtimes L(\lambda^r), \qquad \dim L(\lambda) = \prod_{k=1}^{r} \dim L(\lambda^k).$$

Every finite-dimensional simple $\mathfrak{g}$-module has this form.

Proof. Let T be the outer tensor product, and let v_k be a highest weight vector of $L(\lambda^k)$. A tensor of weight vectors of weights μ^k has weight $(\mu^1, \dots, \mu^r)$. This equals λ only if every $\mu^k = \lambda^k$, so

$$T_\lambda = L(\lambda^1)_{\lambda^1} \otimes \cdots \otimes L(\lambda^r)_{\lambda^r} = \mathbb{C}\, v, \qquad v = v_1 \otimes \cdots \otimes v_r.$$

For $y_k \in U(\mathfrak{g}_k)$, the product $y_1 \cdots y_r$ acts by

$$(y_1 \cdots y_r)v = \rho_1(y_1)v_1 \otimes \cdots \otimes \rho_r(y_r)v_r.$$

Since $U(\mathfrak{g}_k)v_k = L(\lambda^k)$, the vector v generates T. The module T is finite-dimensional by Theorem 13.9 for each $\mathfrak{g}_k$, and it is the direct sum of its weight spaces. So Lemma 13.8 shows that T is simple.

Also $\mathfrak{n}_+ = \bigoplus_k \mathfrak{n}_+^{(k)}$ kills v. By (F2), T is a simple quotient of $M(\lambda)$, so $T \cong L(\lambda)$. The last sentence follows from Theorem 13.10. ∎

13.5 Example: the fundamental $\mathfrak{sl}_3$ module

For $\mathfrak{g} = \mathfrak{sl}_3$ take

$$h_1 = E_{11} - E_{22}, \quad e_1 = E_{12}, \quad f_1 = E_{21},$$
$$h_2 = E_{22} - E_{33}, \quad e_2 = E_{23}, \quad f_2 = E_{32}.$$

We record a weight by its values $(\mu(h_1), \mu(h_2))$. The brackets $[h_i, e_j] = \alpha_j(h_i)e_j$ give $\alpha_1 = (2, -1)$ and $\alpha_2 = (-1, 2)$.

The weight ω_1. Here $(m_1, m_2) = (1, 0)$. With $v = v_{\omega_1}$,

$$V(\omega_1) = M(\omega_1) \big/ \big(U(\mathfrak{g}) f_1^2 v + U(\mathfrak{g}) f_2 v\big).$$

A dominant weight $\mu = \omega_1 - a\alpha_1 - b\alpha_2$ with $a, b \in \mathbb{Z}_{\geq 0}$ satisfies

$$\mu(h_1) = 1 - 2a + b \geq 0, \qquad \mu(h_2) = a - 2b \geq 0.$$

The two inequalities give

$$4b \le 2a \le 1 + b, \qquad \text{so} \qquad 3b \le 1, \quad b = 0, \quad a = 0.$$

The only dominant weight $\mu \le \omega_1$ is ω_1. Simple reflections give

$$s_1\omega_1 = (1,0) - \alpha_1 = (-1,1) = \omega_1 - \alpha_1,$$
$$s_2(-1,1) = (-1,1) - \alpha_2 = (0,-1) = \omega_1 - \alpha_1 - \alpha_2.$$

So the set

$$S = \{(1,0), (-1,1), (0,-1)\}$$

lies in $W\omega_1$. Moreover, s_1 swaps the first two elements and fixes $(0,-1)$. Also s_2 swaps the last two and fixes $(1,0)$. So S is W-stable by (F3), and $S = W\omega_1$. By the proof of Theorem 13.9, every weight of $V(\omega_1)$ lies in this orbit. Proposition 13.7 and $\dim V(\omega_1)_{\omega_1} = 1$ give $\dim V(\omega_1) = 3$.

Compare with $\mathbb{C}^3$ and its standard basis (b_1, b_2, b_3). The diagonal matrices

$$D(h_1) = \operatorname{diag}(1,-1,0), \qquad D(h_2) = \operatorname{diag}(0,1,-1)$$

give the three weights above:

$$b_1 : (1,0), \qquad b_2 : (-1,1), \qquad b_3 : (0,-1).$$

Also $D(e_1)b_1 = D(e_2)b_1 = 0$. The lowering matrices are

$$D(f_1) = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad D(f_2) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

They give

$$D(f_1)b_1 = b_2, \qquad D(f_1)^2 b_1 = D(f_1)b_2 = 0,$$
$$D(f_2)b_1 = 0, \qquad D(f_2)D(f_1)b_1 = b_3.$$

So b_1 satisfies the defining relations. Also b_2 and b_3 lie in the submodule generated by b_1, so b_1 generates $\mathbb{C}^3$. By (F2), $v \mapsto b_1$ gives a surjection $M(\omega_1) \to \mathbb{C}^3$ killing $f_1^2 v$ and $f_2 v$, hence $N(\omega_1)$. The induced surjection $V(\omega_1) \to \mathbb{C}^3$ is an isomorphism by dimension. It sends u, f_1u, f_2f_1u to b_1, b_2, b_3.

Self-checks

(1) For $\mathfrak{g} = \mathfrak{sl}_2$ and $\lambda = m\omega_1$ with $m \in \mathbb{Z}_{\ge 0}$, describe $V(\lambda)$.

(2) For $\mathfrak{sl}_3$, is $2\omega_1 - 3\alpha_1$ a weight of $L(2\omega_1)$?

(3) For $\mathfrak{sl}_2$ and $\lambda = -\omega_1$, which step fails? Is $L(\lambda)$ finite-dimensional?

Answers. (1) $N(\lambda)$ is spanned by the $f^k v_\lambda$ with $k > m$, so $V(\lambda) \cong V_m$. (2) No. Its h_1-value is $2 - 6 = -4$, so s_1 sends it to $2\omega_1 + \alpha_1$, which is not $\le 2\omega_1$. Weyl invariance excludes it. (3) The definition of $N(\lambda)$ needs $\lambda(h) \in \mathbb{Z}_{\ge 0}$, but $\lambda(h) = -1$. For the image $\bar{v}$ of v_λ in $L(\lambda)$, Lemma 13.1 gives $ef^k\bar{v} = -k^2 f^{k-1}\bar{v}$ for $k \ge 1$. If some $f^k\bar{v}$ vanished, the least such $k \ge 1$ would give $0 = -k^2 f^{k-1}\bar{v} \ne 0$. So no $f^k\bar{v}$ vanishes. Their weights differ, so they are linearly independent, and $L(\lambda)$ is infinite-dimensional.

Summary

For $\lambda \in P_+$, the finite-dimensional simple module $L(\lambda)$ is the quotient of $M(\lambda)$ by the submodule generated by $f_i^{m_i+1} v_\lambda$. Thus P_+ classifies the finite-dimensional simple modules up to isomorphism. For a sum of simple ideals, these modules are outer tensor products of simple modules of the ideals.

Reading

Humphreys, *Introduction to Lie Algebras and Representation Theory*, Section 21. Kirillov, *An Introduction to Lie Groups and Lie Algebras*, Sections 8.3 and 8.9. Carter, *Lie Algebras of Finite and Affine Type*, Section 10.3.